

Kitchenware object classification

Course Aprendizaje Automático Avanzado

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1. Introduction

Kitchenware object classification

Final project - Advanced Machine Learning 2025-2026

- Group 02

Part A – Architectures Samuel

Part B – Results Guillermo

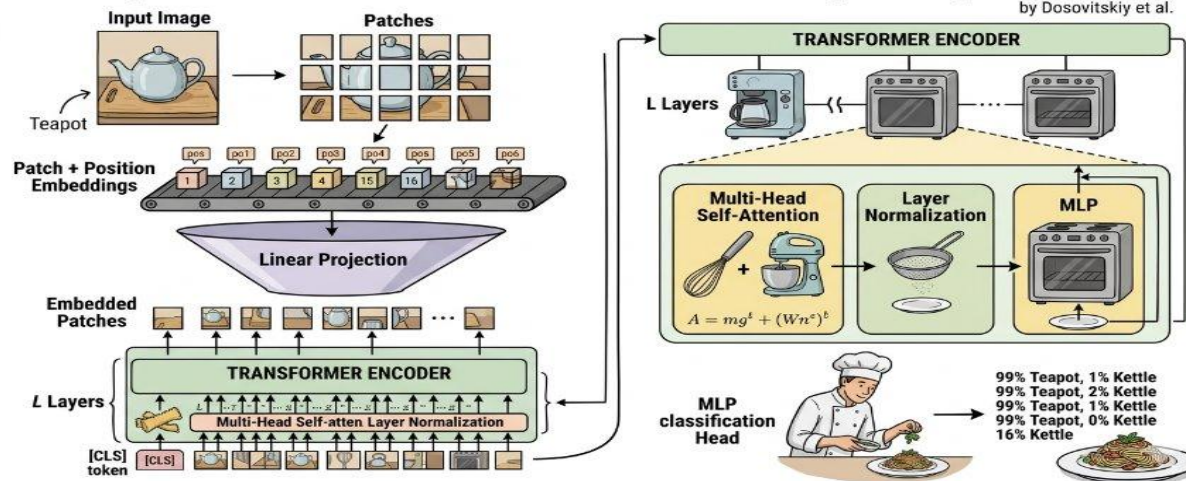
Part C – Strategies Álvaro

2. Vision Transformer (ViT) vs. Baseline CNN

- Paradigm Shift
- Core Mechanism:
Attention Heads
- The Impact on
Kitchenware

An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

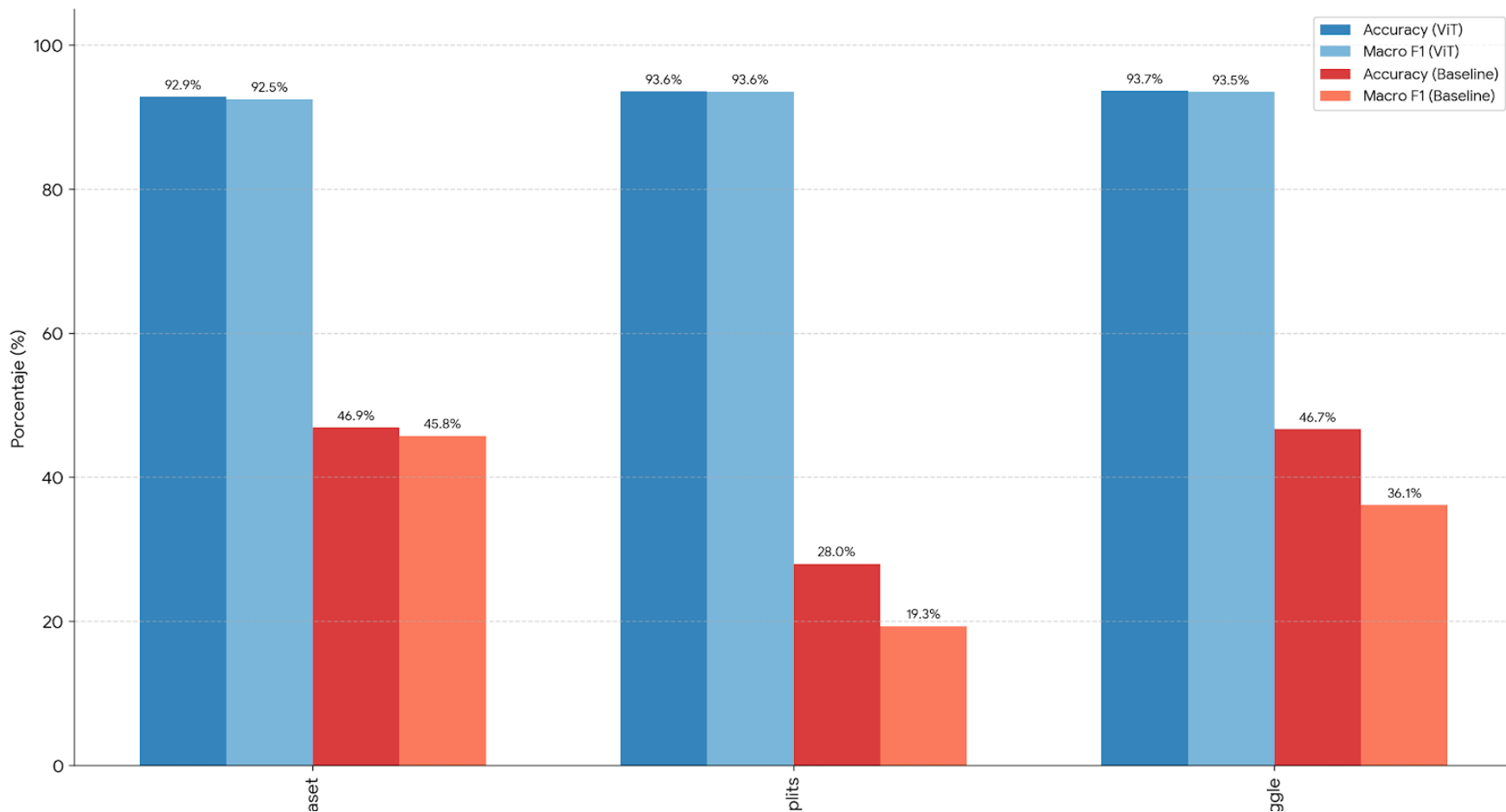
by Dosovitskiy et al.



3. Performance Comparison (The Leap in Quality)

-> Impact on Complex Environments

Comparación de Rendimiento Global: ViT_B_16 vs Baseline por Dataset



4. Limitations: The Cost of Performance

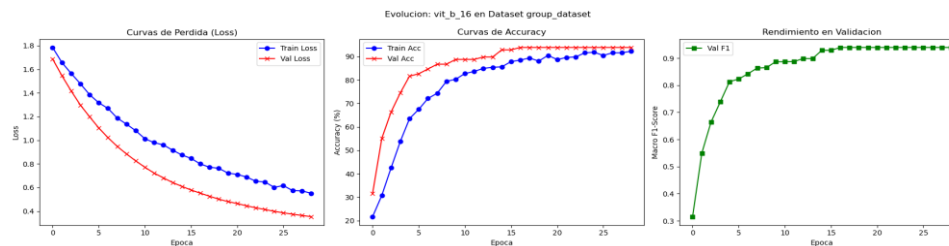
When ViT Is Not the Optimal Choice

"Data-Hungry" Nature:

Lacks inductive biases; requires massive data volumes and risks overfitting on small datasets.

High Computational Cost:

Demands powerful hardware infrastructure for both training and real-time inference.

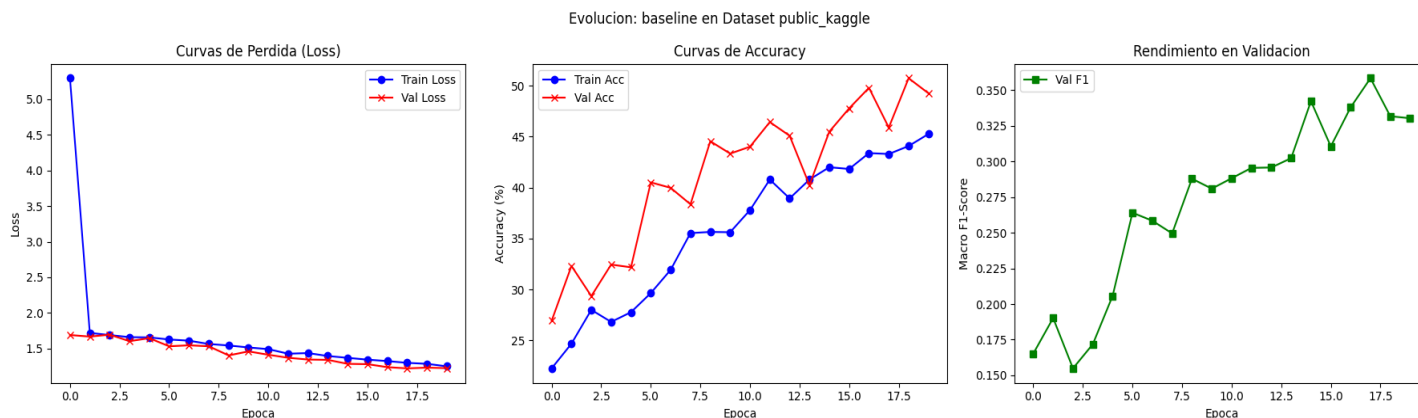


Vit->
group_dataset
(600 images)

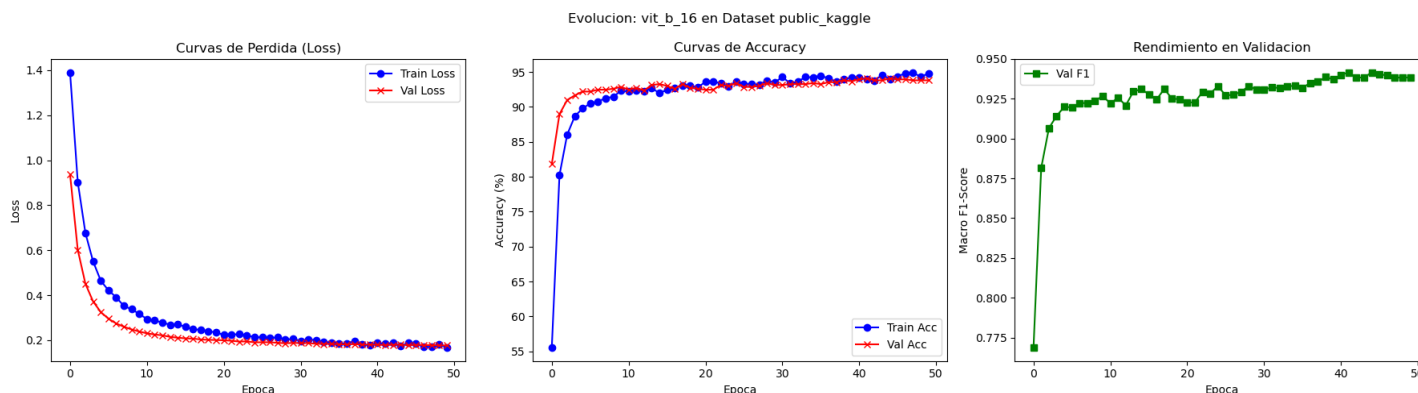
5. Training Diagnostics: Learning Curves

- Baseline vs. ViT-B-16 Convergence

1. [Baseline (CNN)]



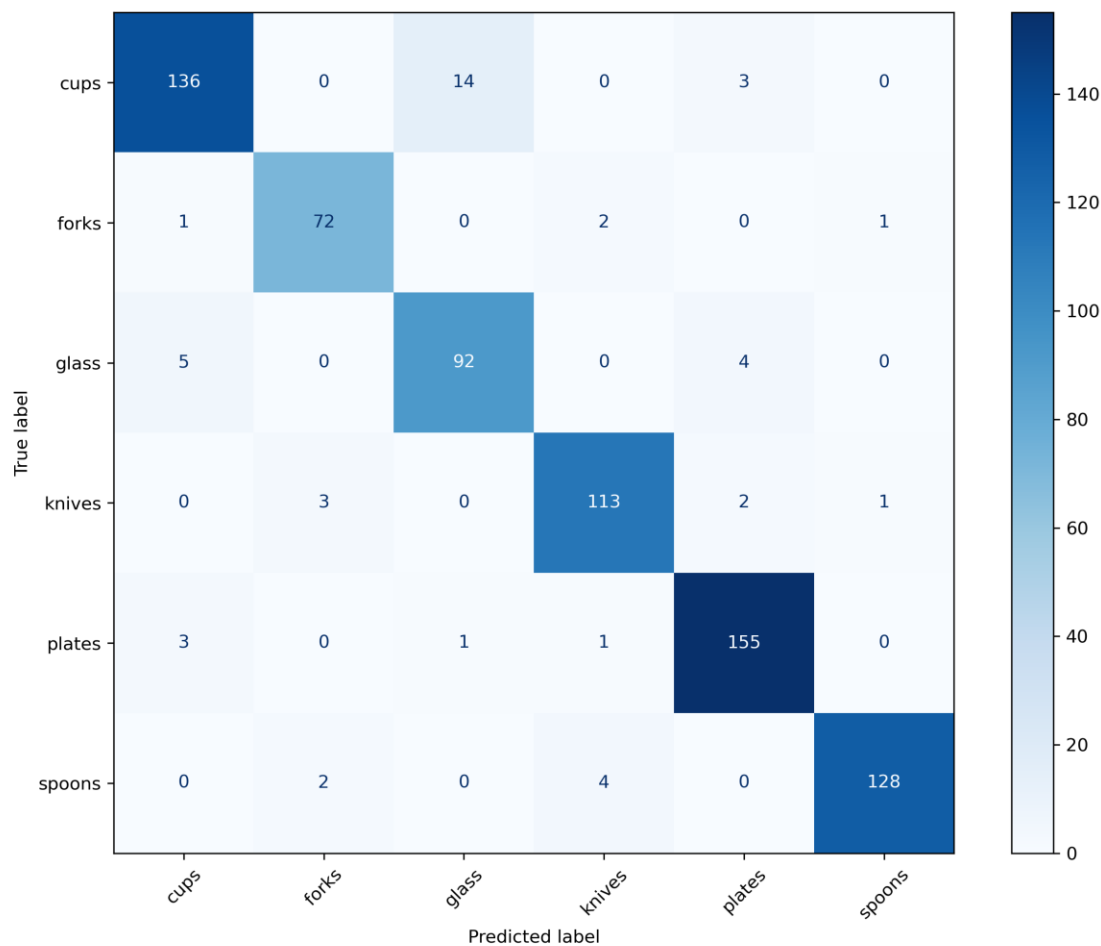
2. [ViT-B-16 Model]



6. Inter-Class Confusion Patterns

- Fine-Grained Error Analysis for ViT-B-16

Matriz de Confusión - ViT_B_16 (public_kaggle)



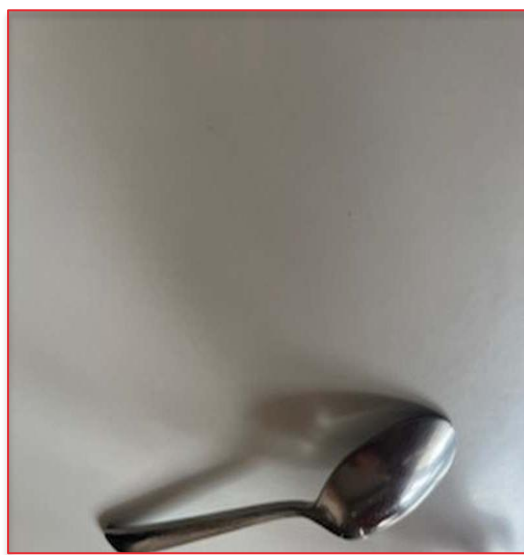
1. Regarding conflict identification
2. Regarding the justification of this error

7. Qualitative Analysis: Misclassified Images

Misclassified Images(Group_dataset) - ViT



- Ground Truth: **Knife**
- ViT Prediction: **Spoon**



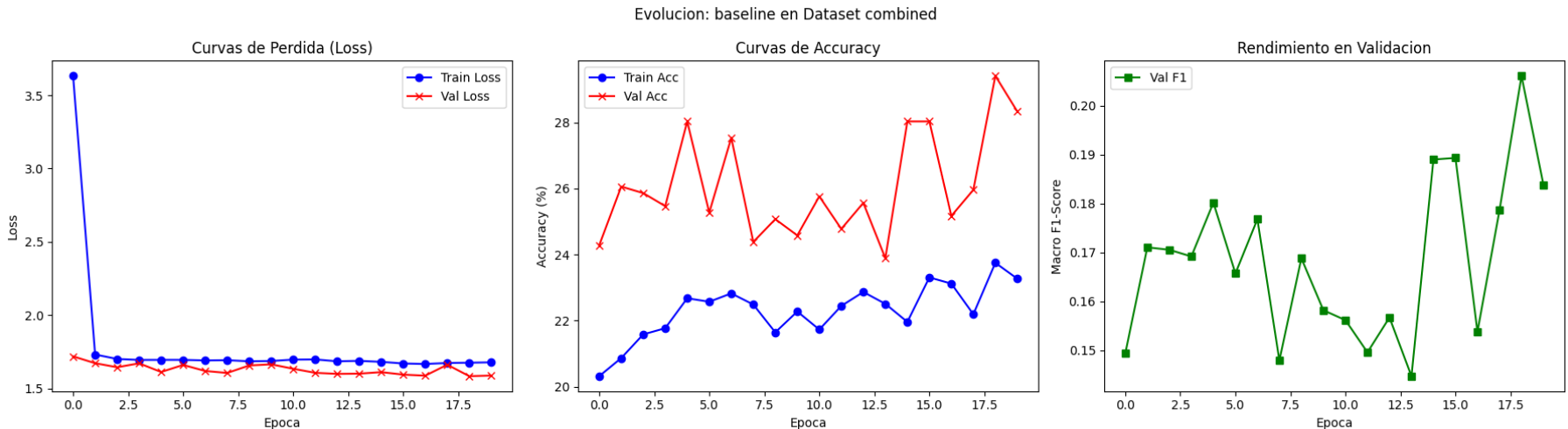
- Ground Truth: **Spoon**
- ViT Prediction: **Fork**



- Ground Truth: **Cup**
- ViT Prediction: **Glass**

8. The Most Critical Scenario: Combined Dataset

Class	Baseline F1-Score	ViT-B/16 F1-Score	Dataset Diagnosis
Forks	0.00%	93.00%	Critical imbalance and occlusion
Glasses	0.00%	91.00%	Transparencies and variable backgrounds
Spoons	0.00%	95.35%	Extreme geometric similarity
Global	19.33%	93.57%	Evidence of Maximum Difficulty



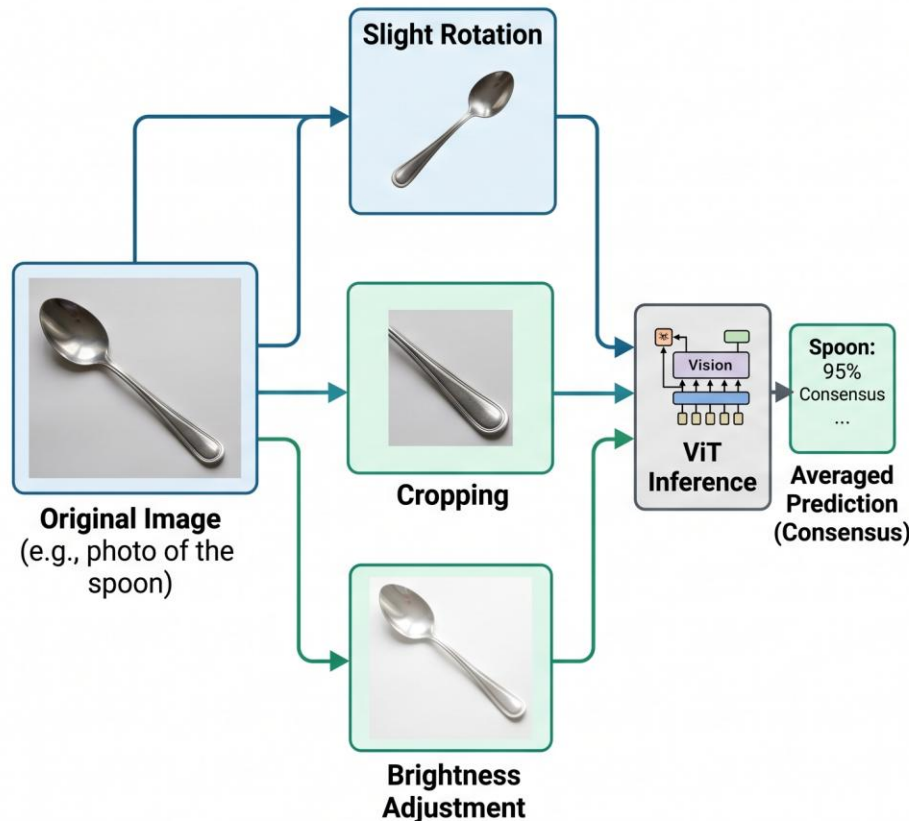
9. Improvement Strategies: The Ablation Study

Samuel	Álvaro	Guillermo
<u>Optimization</u>	<u>Loss Function</u>	<u>Regularization</u>
Transition to AdamW + Learning Rate Scheduler. Key to avoiding local minima by dynamically adjusting the learning rate step	Restructuring feature extraction in dense layers. Aimed to penalize errors more heavily in minority classes.	Dropout (0.3/0.4) and BatchNorm to force the network not to rely on a single pixel. Useful in small datasets to prevent memorization, but insufficient against complex backgrounds.

10. Strategy Impact by Dataset (F1-Score)

Evaluated Model	Key Strategy	F1-Score (Kaggle)	F1-Score (Combined)	F1-Score (Group_dataset)
Baseline CNN	None	36.14%	19.33%	45.77%
Improv_Álvaro	Feature Restructuring	44.21%	47.07%	55.04%
Improv_Guille	Dropout + BatchNorm	51.06%	46.60%	60.53%
Improv_Samuel	AdamW + LR Scheduler	73.98%	66.60%	65.60%
ViT-B/16 (Reference)	Self-Attention	93.51%	93.57%	92.08%

11. Recommendation for future improvements stated.



Implementation of Test-Time Augmentation (TTA)

- **The Current Problem:** Vulnerability to partial occlusions and ambiguous perspectives (Spoon/Fork confusion).
- **The TTA Concept:** Applying slight transformations to the image during inference, not just during training.
- **The Expected Impact:** By averaging predictions from multiple views of the same object, the model 'votes' and drastically reduces the error margin caused by noise or reflections.

12. Lessons Learned

Stagnation in Local Minima due to Visual Noise

The Biggest Challenge

- **Early Learning Stall:** CNN baseline stuck at 36.14% F1.
- **Chaotic Loss Landscape:** Driven by metal reflections and backgrounds.
- **Optimizer Failure:** Trapped permanently in local minima.
- **Critical Impact:** 0.00% F1-score on forks.

How Resolved

- **Dynamic Optimization:** Switched strictly to AdamW optimizer.
- **Weight Decay Separation:** Prevented model from memorizing noise.
- **Smart Scheduling:** Implemented ReduceLROnPlateau tool.
- **The Result:** Doubled performance to 73.98% F1

**Thank you for your
attention**